DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

Software Engineering - Machine Learning

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN Department of Artificial Intelligence and Machine Learning

Submitted by:

Aditya Kumar Choubey – 1DS20AI004

Conducted at

Harman Connected Services

Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78
2023-24

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

CERTIFICATE

This is to certify that the Internship titled **Machine Learning Intern** carried out by **Aditya Kumar Choubey** bearing USN **1DS20AI004**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

X

Kaushlesh Kumar
Director Engineering, SW

Signature of Reporting Head
Kaushlesh Kumar
Director Software Engineering
Harman

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner	Signature with Date
1) ________________	________________
2) ________________	________________

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning

DECLARATION

I, **Aditya Kumar Choubey,** final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **Harman Connected Services**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023-2024.

Date: 1-05-2024 USN: 1DS20AI004

Place: Bangalore NAME:Aditya Kumar Choubey

OFFER LETTER

HARMAN

Harman Connected Services Corporation India Pvt. Ltd
No.3 & 3A, EOIZ Industrial Area, Survey No. 85 & 86
Sadaramangala Village, Krishnarajapuram Hobli
Bangalore - 560066, Karnataka, India
+91(80)3737 1000
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HARMAN
A SAMSUNG COMPANY

29 January 2024

Aditya Choubey
NA
Bengaluru- NA
Karnataka

Sub: Offer of Internship

Hello Aditya,

Congratulations! Pursuant to our discussion, we are delighted to offer you employment opportunity with **Harman Connected Services Corp. India Pvt. Ltd.,Bangalore**. Your initial place of posting will be **Harman India** situated at **IN_Bangalore_EOIZ Indust Area Campus_HCS,Bangalore, Karnataka**. In this capacity you will report to **Business Unit/Function Head** or his/her nominee.

Further, the "Company" for all purposes in this Offer of Internship (**"Offer"**) shall mean and include the Affiliates of the Company, where the context may so require.

Following are the terms and conditions associated with your Internship:

1. Definitions:

"**Company**" means **Harman Connected Services Corp. India Pvt. Ltd.**

"**Affiliate**" shall mean, with respect to the Company, any individual, corporation, association or other Person (*defined as below*), which directly or indirectly, controls the Company, is controlled by the Company or is under common control with the Company. For this purpose, "control" (including, with its correlative meanings, "controlled by" and "under common control with") shall mean the possession, directly or indirectly, of the power to direct or cause the direction of management or policies of a Person, whether through the ownership of securities or partnership or other ownership interests, by contract or otherwise.

"**Person**" means an individual, company, association, joint venture, partnership, estate, trust, an association of persons and any other entity or organization, governmental or otherwise.

2. **Date of Joining:** Your Internship with the Company shall commence on **1 February 2024**. In the event you fail to report at the Company location mentioned above on the mentioned date, the offer made herein shall stand withdrawn, unless change in the reporting date is communicated and agreed with the Company in writing.

Aditya M. Choubey

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Kaushlesh Kumar Director, Engineering SW, for guiding me during the period of internship.

We express our deep and profound gratitude to our Internship Coordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of the AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Aditya Kumar Choubey
1DS20AI004

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CHAPTER 1

INTRODUCTION

During my tenure as a Machine Learning Intern at Harman Connected Services Inc. from February 1st, 2024, to July 31st, 2024, I undertook a transformative project aimed at revolutionizing the management and optimization of Large Language Models (LLMs). This endeavor was centered on the development of an innovative LLMOps (Large Language Model Operations) framework, with the primary objective of enhancing the efficiency, scalability, and reliability of LLM operations across the organization.

The significance of Large Language Models in various industries, including natural language processing, conversational AI, and content generation, cannot be overstated. Recognizing these challenges as opportunities for improvement, the internship project sought to construct a comprehensive LLMOps framework capable of addressing these complexities and streamlining the operational processes associated with LLMs. The LLMOps framework aimed to provide a standardized and automated pathway for LLM operations, ensuring consistency, quality, and alignment with the business objectives of Harman Connected Services Inc.

By leveraging cutting-edge technologies and methodologies, the framework aimed to optimize resource utilization, minimize deployment time, and maximize model performance. Furthermore, the project emphasized the importance of documentation, testing, and stakeholder engagement to validate the effectiveness and robustness of the LLMOps framework.

This internship project represents not only a technical endeavor but also a strategic initiative to drive innovation and excellence in machine learning operations within the organization. The development of the LLMOps framework signifies a commitment to continuous improvement and a proactive approach to addressing the evolving needs and challenges of the industry. In this report, I will provide a comprehensive overview of the internship project, detailing the project's objectives, methodologies, key deliverables, and expected outcomes. Through this documentation, I aim to showcase the significance of the LLMOps framework in advancing the field of machine learning operations and its potential impact on the future success of Harman Connected Services Inc.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Harman Connected Services, often abbreviated to HCS, is an American software company which is a subsidiary of Samsung Electronics through Harman International Industries. The Connected Services Division supplies software services to the mobile communications industry.

Harman has a workforce of approximately 30,000 people across the Americas, Europe, and Asia. On January 22, 2015, Harman acquired Symphony Teleca from the Symphony Technology Group. The deal was valued at US$780 million. Symphony Teleca was subsequently integrated and rebranded as Harman Connected Services and, in March 2017, Harman became a wholly owned subsidiary of Samsung Electronics.

Key Features:

- **Mission and Values:** At the core of Harman Connected Services Inc. lies a steadfast commitment to excellence, innovation, and customer-centricity. The organization's mission revolves around leveraging cutting-edge technologies to empower businesses and individuals, fostering connectivity, collaboration, and efficiency.
- **Expertise and Offerings:** Harman Connected Services Inc. boasts a diverse portfolio of offerings, encompassing a wide range of technology solutions and services. From software development and engineering to IoT (Internet of Things) solutions and cloud services, the organization's expertise spans across multiple domains, catering to the evolving needs of modern businesses.
- **Industry Impact:** With a strong foothold in the global market, Harman Connected Services Inc. has made significant contributions to various industries, including automotive, healthcare, retail, and telecommunications. By harnessing the power of technology, the organization has enabled businesses to enhance operational efficiency, improve decision-making, and drive sustainable growth.

CHAPTER 3

SCOPE OF WORK

The scope of work for developing the LLMOps (Large Language Model Operations) framework encompasses a multifaceted approach aimed at addressing the complexities and challenges associated with the lifecycle of Large Language Models (LLMs). Over the course of the internship project at Harman Connected Services Inc., the following key activities will be undertaken:

1.**Research and Analysis:** The initial phase will involve conducting comprehensive research to gain a deep understanding of the complexities and challenges inherent in the life cycle of LLMs.

2. **Toolchain Evaluation:** A critical aspect of the project will be the evaluation and selection of the appropriate toolchain for LLMOps. This will involve assessing various tools and technologies available in the market to construct a robust framework capable of meeting the organization's specific requirements and objectives.

3. **Framework Design and Development:** The heart of the project lies in designing and developing the LLMOps framework itself. This phase will involve conceptualizing and structuring a comprehensive system that can standardize and automate LLM operations, encompassing everything from model development and training to deployment and monitoring.

4. **Alignment with Business Goals:** Throughout the development process, a key focus will be on ensuring alignment with the overarching business goals and objectives of Harman Connected Services Inc. The LLMOps framework must not only enhance technical processes but also contribute to the organization's strategic initiatives and long-term vision.

5. **Testing and Validation**: Once the LLMOps framework is developed, it will undergo rigorous testing and validation to ensure its effectiveness, reliability, and robustness in real-world scenarios. This will involve conducting various tests, debugging any issues that arise, and iterating on the framework to optimize performance.

6. **Documentation and Presentation**: Comprehensive documentation will be compiled throughout the development process, detailing insights gained, challenges encountered, and solutions implemented. The final LLMOps framework will be presented to project stakeholders, showcasing its functionality, benefits, and potential impact on the organization.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

During the internship period at Harman Connected Services Inc., the company has outlined the following learning objectives aimed at enhancing professional development and maximizing the learning experience:

1. **Deep Understanding of LLM Lifecycle**: One of the primary learning objectives is to gain a comprehensive understanding of the lifecycle of Large Language Models (LLMs). This involves exploring the complexities and challenges associated with LLM operations, from development and training to deployment and monitoring.
2. **Research and Analysis Skills:** The internship provides an opportunity to hone research and analysis skills by delving into industry best practices, analyzing existing frameworks, and identifying areas for improvement in LLM operations. This includes gathering relevant data, synthesizing information, and drawing insights to inform decision-making.
3. **Technical Proficiency in LLMOps:** A key focus of the internship is to develop technical proficiency in LLMOps (Large Language Model Operations). This entails gaining hands-on experience with tools, technologies, and methodologies used in constructing and implementing the LLMOps framework, including model development, deployment pipelines, and monitoring systems.
4. **Business Alignment and Communication:** Understanding the alignment between technical solutions and business objectives is crucial. The internship provides an opportunity to enhance communication skills by effectively articulating technical concepts to non-technical stakeholders and aligning the LLMOps framework with the strategic goals of Harman Connected Services Inc.
5. **Testing and Validation Techniques:** The internship involves learning about testing and validation techniques to ensure the effectiveness, reliability, and robustness of the LLMOps framework. This includes conducting various tests, debugging issues, and iterating on the framework to optimize performance and meet quality standards.
6. **Documentation and Presentation Skills:** Learning to document the development process, insights gained, and challenges encountered will be emphasized, along with presenting the final LLMOps framework to project stakeholders in a clear and concise manner.

CHAPTER 5

CHALLENGES AND SOLUTION

CHALLENGES

Developing the LLMOps framework at Harman Connected Services Inc. presented several challenges, each demanding innovative solutions to ensure project success:

1: **Complexity of LLM Lifecycle** Understanding the intricate lifecycle of Large Language Models (LLMs) proved challenging. The diverse stages, from data preprocessing to deployment and monitoring, required deep comprehension and management.

2: **Toolchain Selection** Selecting the right toolchain for LLMOps amidst a myriad of options was daunting. We needed tools that aligned with our scalability, integration, and support requirements.

3: **Business Alignment** Ensuring the LLMOps framework aligned with Harman Connected Services Inc.'s business objectives posed a significant challenge. Demonstrating its tangible value and contribution to strategic goals was imperative.

SOLUTION

Extensive research and collaboration were pivotal. We delved into industry practices, held regular meetings with experts, and conducted thorough analyses. This enabled us to design a robust LLMOps framework that addressed the complexities of each lifecycle stage.

We adopted a meticulous evaluation process, prototyping and testing various tool combinations. This allowed us to identify the optimal toolchain configuration that best suited our framework's needs.

Close collaboration with stakeholders and effective communication channels were key. We actively sought feedback and incorporated stakeholder requirements into our framework design. Prototypes and simulations were instrumental in illustrating the framework's alignment with business goals..

Effective Time Management: Prioritize tasks and allocate time efficiently to ensure progress and meet deadlines.

Continuous Learning: Stay updated with the latest advancements in LLM through ongoing learning and exploration.

CHAPTER 6
ACHIEVEMENTS AND CONTRIBUTION

During my internship at Harman Connected Services Inc., my involvement in developing the LLMOps (Large Language Model Operations) framework led to several notable achievements and contributions:

1. **Designing a Robust LLMOps Framework**: I played a pivotal role in designing and developing a comprehensive LLMOps framework that streamlined the entire lifecycle of Large Language Models (LLMs). Through meticulous research, analysis, and collaboration with the team, we crafted a solution that addressed the complexities and challenges associated with LLM operations.
2. **Toolchain Selection and Integration:** I contributed to the evaluation and selection of the optimal toolchain for LLMOps, ensuring scalability, flexibility, and compatibility with the organization's requirements. By integrating various tools and technologies seamlessly, we created a cohesive framework that enhanced operational efficiency and effectiveness.
3. **Testing and Validation:** I played a key role in testing and validating the LLMOps framework to ensure its reliability and robustness in real-world scenarios. Through rigorous testing procedures and meticulous validation processes, we identified and addressed potential issues, ensuring the framework's performance met quality standards and user expectations.
4. **Documentation and Knowledge Sharing**: I contributed to documenting the development process, insights gained, and solutions implemented throughout the project. By maintaining comprehensive documentation, I facilitated knowledge sharing within the team and ensured continuity in the framework's maintenance and evolution.
5. **Presentation to Stakeholders**: I participated in presenting the final LLMOps framework to project stakeholders, showcasing its functionality, benefits, and potential impact on the organization. Through clear and concise presentations, I effectively communicated the framework's value proposition, garnering support and enthusiasm from stakeholders.

CHAPTER 7

REFLECTION AND ANALYSIS

My internship experience at Harman Connected Services Inc., particularly my involvement in developing the LLMOps (Large Language Model Operations) framework, has been profoundly enriching and transformative. Through reflection and analysis, I have gained valuable insights and lessons that have shaped my personal and professional growth.

The project provided me with an invaluable opportunity to deepen my technical proficiency in machine learning operations. From understanding the intricacies of the LLM lifecycle to evaluating and integrating diverse toolchains, I expanded my knowledge and skill set significantly.

Collaborating with cross-functional teams and stakeholders was instrumental in the success of the project. Effective communication, both verbal and written, played a crucial role in aligning the LLMOps framework with the organization's business goals and gathering feedback throughout the development process. Learning to navigate diverse perspectives, resolve conflicts, and foster collaboration has been invaluable in enhancing my interpersonal skills and teamwork abilities.

The project presented various challenges, from understanding the complexities of the LLM lifecycle to selecting the optimal toolchain and ensuring alignment with business objectives. Each challenge demanded innovative solutions, creativity, and adaptability. Embracing a problem-solving mindset and being open to experimentation and iteration were essential in overcoming obstacles and achieving our objectives.

Managing a complex project with multiple stakeholders and deadlines required effective project management and time management skills. Prioritizing tasks, setting achievable goals, and maintaining momentum were crucial in staying on track and delivering results. The experience honed my ability to manage resources efficiently and navigate project complexities effectively.

The internship experience has instilled in me a passion for continuous learning and growth. Reflecting on my experiences, identifying areas for improvement, and seeking opportunities for further development have become integral parts of my professional journey. Embracing a growth mindset and remaining curious and adaptable are key principles that I will carry forward into future endeavors.

CHAPTER 8

RECOMMENDATIONS

Based on my internship experience at Harman Connected Services Inc. and my involvement in developing the LLMOps framework, I would like to offer some recommendations for consideration such as encouraging a culture of continuous improvement within the organization by fostering an environment where innovation and experimentation are valued.

Promote cross-functional collaboration and communication across different departments and teams within the organization. Encourage interdisciplinary teamwork to leverage diverse perspectives and expertise in addressing complex challenges and driving innovation in machine learning operations.

Emphasize the importance of comprehensive documentation and knowledge sharing practices to capture insights, lessons learned, and best practices from projects such as the development of the LLMOps framework. Establish repositories and platforms for sharing documentation, code, and resources to facilitate collaboration and learning across the organization.

Prioritize stakeholder engagement throughout the project lifecycle by actively involving key stakeholders in decision-making processes and seeking their input and feedback. Ensure that the LLMOps framework aligns with the strategic objectives and priorities of the organization and address any concerns or requirements raised by stakeholders in a timely manner.

Invest in technology and infrastructure to support the implementation and scaling of the LLMOps framework and other machine learning initiatives. Provide access to state-of-the-art tools, platforms, and resources to empower employees to innovate and optimize machine learning operations effectively.

Establish key performance indicators (KPIs) and metrics to measure the effectiveness and impact of the LLMOps framework on business outcomes. Regularly monitor and evaluate performance against these metrics to identify areas for improvement and optimization and ensure that the framework continues to deliver value to the organization. By implementing these recommendations, Harman Connected Services Inc. can further strengthen its capabilities in machine learning operations and drive continued innovation and excellence in this critical area.

CHAPTER 9

CONCLUSION

In conclusion, my internship journey at Harman Connected Services Inc. has been an immensely rewarding experience, marked by significant learning, growth, and contribution. Through my involvement in developing the LLMOps framework, I have gained valuable insights into the complexities and challenges of machine learning operations, honed my technical skills, and strengthened my ability to collaborate effectively in a dynamic and interdisciplinary environment.

The project provided me with the opportunity to apply theoretical knowledge to real-world scenarios, navigate complex challenges, and deliver tangible results that align with the organization's strategic objectives.

Moreover, the experience has underscored the importance of continuous learning, adaptability, and innovation in driving meaningful change and staying ahead in a rapidly evolving technological landscape.

As I reflect on my journey, I am grateful for the mentorship, support, and opportunities provided by Harman Connected Services Inc. and look forward to applying the lessons learned and experiences gained to future endeavors.

Moving forward, I am committed to leveraging my newfound skills and insights to make meaningful contributions to the field of machine learning and beyond, while continuing to pursue excellence and growth in my professional journey.

CHAPTER 10

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